THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 150	Find the new time first, then the angle: each hour mark is 30°.
2	**(a)** 1170	$360n + 90$.
3	**(a)** 180	Net turn.
4	**(a)** 30	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
5	**(a)** 25	The minute hand turns 6° per minute.
6	**(c)** 4	Each line gives one straight angle.
7	**(d)** 72	$360^\circ \div 5$.
8	**(d)** 150	Multiply the angle between neighbours by the number of gaps.
9	**(a)** Both A and R are true and R is the correct explanation of A.	135 lies between 90 and 180, and R is exactly the definition.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the value of a right angle does not explain the value of a straight angle.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hour hand points to 4 on the dial; smaller angle $= 120^\circ$ (1); Hour hand turns $2 \times 30 = 60^\circ$ in 2 hours (1).	1+1
12	Total $= 360 \times 2 + 90 = 810^\circ$ (1); Right angles $= (810) \div 90 = 9$ (1).	1+1
13	Each pair of rays makes one angle (1); $\frac{8 \times 7}{2} = 28$ (1).	1+1
14	Hour hand points to 6 on the dial; smaller angle $= 180^\circ$ (1); Hour hand turns $2 \times 30 = 60^\circ$ in 2 hours (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hour hand points to 1 on the dial; smaller angle $= 30^\circ$ (1); Larger angle $= 360 - 30 = 330^\circ$ (1); Reflex angle: the larger angle 330° (1).	1+1+1
16	Minutes $= 90 \div 6 = 15$ (1); Right angles $= 90 \div 90 = 1$ (1); Time shown: 2:15 (1).	1+1+1
17	Slices $= 2 \times 2 = 4$ (1); Angle $= 360 \div 4 = 90^\circ$ (1); Type: right (1).	1+1+1
	OR: Hour hand points to 5 on the dial; smaller angle $= 150^\circ$ (1); Larger angle $= 360 - 150 = 210^\circ$ (1); Reflex angle: the larger angle 210° (1).	1+1+1
18	Minutes $= 180 \div 6 = 30$ (1); Right angles $= 180 \div 90 = 2$ (1); Time shown: 4:30 (1).	1+1+1
	OR: Slices $= 2 \times 3 = 6$ (1); Angle $= 360 \div 6 = 60^\circ$ (1); Type: acute (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $20 \times 6 = 120^\circ$ (1). (ii) 30° (1). (iii) At 4 o'clock the angle is $4 \times 30 = 120^\circ$, an obtuse angle (1). (iv) At 7 o'clock one angle is 210° and the smaller angle is $360^\circ - 210^\circ = 150^\circ$ (1); the 210° angle is a reflex angle (1).	1+1+1+2
20	(a) Hour hand: 4 hour marks of $30^\circ = 120^\circ$ (1); (b) Minute hand has completed 4 full turns $= 1440^\circ$ and stands at 12 (1); (c) Smaller angle $= 120^\circ$ (1); (d) obtuse angle (2).	1+1+1+2
	OR: Net $= 270 - 90 + 135 = 315^\circ$ (1); Direction 315° clockwise from North (1); To face North again: turn clockwise 45° more (1); Check: $315 + 45 = 360^\circ$ (2).	1+1+1+2

Mistakes to avoid:

- Reading the wrong scale on the protractor (inner vs outer). Decide first whether the angle is acute or obtuse.
- Using 30° per minute for the minute hand; it is 6° per minute, and 30° per hour is for the hour hand.
- Calling a 200° or 225° angle obtuse. Obtuse angles are less than 180°; larger ones are reflex.
- Saying a ray has two end points; a ray has exactly one end point.
- Counting line segments for n points as $n(n-1)$ without dividing by 2 (segment AB and BA are the same).

- Placing the protractor's centre off the vertex, or starting the scale from the wrong arm.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 270	Net turn.
2	**(d)** 60	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
3	**(a)** 40	The minute hand turns 6° per minute.
4	**(a)** 5	Each line gives one straight angle.
5	**(b)** 60	$360^\circ \div 6$.
6	**(c)** 160	Multiply the angle between neighbours by the number of gaps.
7	**(d)** 30	Find the new time first, then the angle: each hour mark is 30°.
8	**(b)** 2610	$360n + 90$.
9	**(c)** A is true but R is false.	It turns 360° in 60 minutes, i.e. 6° per minute, so $15 \times 6 = 90^\circ$. A is true, R is false.
10	**(d)** A is false but R is true.	200° is a reflex angle, so A is false. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total $= 360 \times 3 + 180 = 1260^\circ$ (1); Right angles $= (1260) \div 90 = 14$ (1).	1+1
12	Each pair of rays makes one angle (1); $\frac{9 \times 8}{2} = 36$ (1).	1+1
13	Hour hand points to 7 on the dial; smaller angle $= 150^\circ$ (1); Hour hand turns $2 \times 30 = 60^\circ$ in 2 hours (1).	1+1
14	Total $= 360 \times 4 + 270 = 1710^\circ$ (1); Right angles $= (1710) \div 90 = 19$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hour hand points to 7 on the dial; smaller angle $= 150^\circ$ (1); Larger angle $= 360 - 150 = 210^\circ$ (1); Reflex angle: the larger angle 210° (1).	1+1+1
16	Minutes $= 270 \div 6 = 45$ (1); Right angles $= 270 \div 90 = 3$ (1); Time shown: 5:45 (1).	1+1+1
17	Slices $= 2 \times 4 = 8$ (1); Angle $= 360 \div 8 = 45^\circ$ (1); Type: acute (1).	1+1+1
	OR: Hour hand points to 8 on the dial; smaller angle $= 120^\circ$ (1); Larger angle $= 360 - 120 = 240^\circ$ (1); Reflex angle: the larger angle 240° (1).	1+1+1
18	Minutes $= 360 \div 6 = 60$ (1); Right angles $= 360 \div 90 = 4$ (1); Time shown: 8:00 (1).	1+1+1
	OR: Slices $= 2 \times 5 = 10$ (1); Angle $= 360 \div 10 = 36^\circ$ (1); Type: acute (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{5 \times 4}{2} = 10$ (1). (ii) $\frac{6 \times 5}{2} = 15$ (1). (iii) $15 - 10 = 5$, since F pairs with each of the 5 stops (1). (iv) Each of the 8 stops pairs with 7 others, giving 56, and each segment is counted twice, so $56 \div 2 = 28$ segments (2).	1+1+1+2
20	(a) Hour hand: 7 hour marks of $30^\circ = 210^\circ$ (1); (b) Minute hand has completed 7 full turns $= 2520^\circ$ and stands at 12 (1); (c) Smaller angle $= 150^\circ$ (1); (d) obtuse angle (2).	1+1+1+2
	OR: Net $= 90 - 180 + 225 = 135^\circ$ (1); Direction 135° clockwise from North (1); To face North again: turn clockwise 225° more (1); Check: $135 + 225 = 360^\circ$ (2).	1+1+1+2

Mistakes to avoid:

- Reading the wrong scale on the protractor (inner vs outer). Decide first whether the angle is acute or obtuse.
- Using 30° per minute for the minute hand; it is 6° per minute, and 30° per hour is for the hour hand.
- Calling a 200° or 225° angle obtuse. Obtuse angles are less than 180°; larger ones are reflex.
- Saying a ray has two end points; a ray has exactly one end point.
- Counting line segments for n points as $n(n-1)$ without dividing by 2 (segment AB and BA are the same).

- Placing the protractor's centre off the vertex, or starting the scale from the wrong arm.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 55	The minute hand turns 6° per minute.
2	**(d)** 6	Each line gives one straight angle.
3	**(a)** 45	$360^\circ \div 8$.
4	**(a)** 135	Multiply the angle between neighbours by the number of gaps.
5	**(b)** 120	Find the new time first, then the angle: each hour mark is 30°.
6	**(b)** 3330	$360n + 90$.
7	**(a)** 135	Net turn.
8	**(a)** 150	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Pairs: AB, AC, AD, BC, BD, CD = 6, and R is the reason we count pairs.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the 3 o'clock fact does not explain the 6 o'clock fact.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Each pair of rays makes one angle (1); $\frac{10\times9}{2} = 45$ (1).	1+1
12	Hour hand points to 9 on the dial; smaller angle $= 90^\circ$ (1); Hour hand turns $2 \times 30 = 60^\circ$ in 2 hours (1).	1+1
13	Total $= 360 \times 5 + 90 = 1890^\circ$ (1); Right angles $= (1890) \div 90 = 21$ (1).	1+1
14	Each pair of rays makes one angle (1); $\frac{12\times11}{2} = 66$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hour hand points to 10 on the dial; smaller angle $= 60^\circ$ (1); Larger angle $= 360 - 60 = 300^\circ$ (1); Reflex angle: the larger angle 300° (1).	1+1+1
16	Minutes $= 450 \div 6 = 75$ (1); Right angles $= 450 \div 90 = 5$ (1); Time shown: 9:15 (1).	1+1+1
17	Slices $= 2 \times 6 = 12$ (1); Angle $= 360 \div 12 = 30^\circ$ (1); Type: acute (1).	1+1+1
	OR: Hour hand points to 11 on the dial; smaller angle $= 30^\circ$ (1); Larger angle $= 360 - 30 = 330^\circ$ (1); Reflex angle: the larger angle 330° (1).	1+1+1
18	Minutes $= 540 \div 6 = 90$ (1); Right angles $= 540 \div 90 = 6$ (1); Time shown: 11:30 (1).	1+1+1
	OR: Slices $= 2 \times 9 = 18$ (1); Angle $= 360 \div 18 = 20^\circ$ (1); Type: acute (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Right angle (1). (ii) $90^\circ + 135^\circ = 225^\circ$ (1). (iii) Reflex angle, as $180^\circ < 225^\circ < 360^\circ$ (1). (iv) $360^\circ - 225^\circ = 135^\circ$ more (1); fraction made $= \frac{225}{360} = \frac{5}{8}$ (1).	1+1+1+2
20	(a) Hour hand: 11 hour marks of $30^\circ = 330^\circ$ (1); (b) Minute hand has completed 11 full turns $= 3960^\circ$ and stands at 12 (1); (c) Smaller angle $= 30^\circ$ (1); (d) acute angle (2).	1+1+1+2
	OR: Net $= 315 - 135 + 90 = 270^\circ$ (1); Direction 270° clockwise from North (1); To face North again: turn clockwise 90° more (1); Check: $270 + 90 = 360^\circ$ (2).	1+1+1+2

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- Saying a ray has two end points; a ray has exactly one end point.

- Counting line segments for n points as $n(n-1)$ without dividing by 2 (segment AB and BA are the same).
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